Week 11 — Multi-Omics Paper Analysis Assignment

Instructions: Choose any THREE of the four papers. Answer all three questions for each paper, citing relevant figures, analytical methods, and key results.

Papers selected for this submission: Paper 1, Paper 2 and Paper 3 (three of the four offered). Answers follow each question.

Question	One-line answer (detail in the section below)
Paper 1 — Q1	DIA LC–MS/MS time course in four adipocyte models (days 0–14, n = 3), Spectronaut directDIA at 1% q-value, median-normalised, median CoV 12.3%, permutation ANOVA FDR < 1%.
Paper 1 — Q2	PCA gives the differentiation axis; 1,979 cross-model correlated proteins cluster into early/intermediate/late modules, annotated by Fisher enrichment (BH FDR < 0.1).
Paper 1 — Q3	Proteome vs matched transcriptome shows post-transcriptional control; PCP/SVM localisation plus the lipid-droplet proteome nominate C19orf12, validated by co-IP (TOM40/22/5), knockdown and Seahorse.
Paper 2 — Q1	MiSeq shotgun metagenomes (MetaPhlAn2/HUMAnN2) and UPLC–qTOF metabolomes (MS-DIAL/MS-FINDER) reduced to QC'd feature tables: 310 species, 7,355 pathways, 435 metabolites.
Paper 2 — Q2	DESeq2 + RF Gini selection gives 19 species and 17 metabolites; 10-fold CV AUC 0.91; frozen models reach AUC 0.86–0.95 externally and 0.84–0.94 for cirrhosis vs fibrosis.
Paper 2 — Q3	Species–metabolite networks plus a CDCA culture experiment link taxa to metabolites; adding age, albumin and AST gives staged composite models.
Paper 3 — Q1	208 paired RNA-seq/16S samples processed to VST genes and CLR multi-rank taxa; Procrustes/Mantel show global coupling only in CRC, so integration is within-cohort with meta-analytic combination.
Paper 3 — Q2	Sparse CCA (L1, LOOCV-tuned, FDR < 0.1) for pathway-level components; gene-wise lasso + desparsified inference + stability selection (≥ 60%) for gene–taxon pairs; Fisher enrichment for pathways.
Paper 3 — Q3	Shared pathways pair with disease-specific microbes and shared microbes with disease-specific host genes (Streptococcus–RIPK4 in CRC vs DPEP2 in IBS), implying disease-context-dependent crosstalk.

Table 1. Answer overview: the one-line conclusion of each answer; figures and methods are cited inside the answers.

Paper 1 — Spatiotemporal Proteomics of Human Adipogenesis

Nature Metabolism, 2024 | Proteomics + Transcriptomics + Spatial Proteomics

1. Analysis Workflow: How were proteomic data generated, quality-controlled, quantified, and analyzed across adipocyte differentiation?

Answer — Analysis Workflow

Short answer: a 1-h data-independent acquisition (DIA) LC–MS/MS time course across four human adipocyte models, quantified with Spectronaut at 1% q-value, quality-controlled at the level of peptides, proteins and replicates, then processed in Perseus with permutation-based ANOVA (FDR < 1%).

- **Study design (Fig. 1a; Supplementary Tables 1–3):** four human adipocyte precursor (hAPC) models — SGBS and hAPC (non-immortalised) plus TERT-hAPC and hWA (immortalised); hAPC and TERT-hAPC come from the same donor, so the effect of immortalisation can be separated from donor effects. Proteomes were sampled on days 0, 2, 4, 6, 10, 12 and 14 (n = 3 per time point per model) and compared with primary mature adipocytes (pAC), stromal vascular fraction (SVF) and whole WAT from seven donors as the in vivo reference.
- **Sample preparation and acquisition (Methods 'Proteomics', 'LC–MS/MS'):** 2% SDC lysis, trypsin/LysC digestion at 1:50 overnight, TCEP/chloroacetamide reduction and alkylation, StageTip clean-up, and 500 ng of peptides injected on an Orbitrap Exploris 480 with FAIMS (CV 50) coupled to an EASY-nLC 1200. The DIA method used one MS1 scan (300–1,650 m/z, R = 120,000) followed by 66 unequally spaced MS2 windows (R = 15,000, HCD 30%). DIA was chosen over DDA to obtain higher identification rates over a wider dynamic range with fewer missing values.
- **Search and quantification (Methods 'Proteomic data processing'):** Spectronaut v15 directDIA against the UniProt human database (UP000005640 + additional), trypsin with two missed cleavages, fixed carbamidomethylation, variable methionine oxidation and N-terminal acetylation, 3–6 best N fragment ions per peptide, precursor and protein q-value 1%, and median-based global normalisation. This gave 5,979–7,061 protein groups per cell model and 8,268 in the complete dataset, 86% of them supported by unique peptides (Extended Data Fig. 1e,f).
- **Quality control (Fig. 1b; Extended Data Fig. 1g–j):** the LC–MS signal spanned five orders of magnitude; 98% of peptides had 3–8 data points per peak; 72% of protein coefficients of variation (CoVs) were below 20% and 42% below 10%, with a median CoV of 12.3%; replicate Pearson correlation averaged 0.97; and correlations were high between models at matched time points — the measurement criterion that licenses the word 'universal' later in the paper.
- **Statistics (Methods 'Proteome analysis'):** in Perseus 1.6.15, LFQ values were log2-transformed, proteins filtered for at least two valid values, width-adjusted, and missing

values imputed from a down-shifted normal distribution (downshift 1.8, width 0.3); significantly changed proteins were called by permutation-based ANOVA (FDR < 1%) per model, or by two-sided t-test for two conditions; contaminants and reverse hits were removed. A proteomic-ruler analysis converted intensities into copies per cell as an independent normalisation (Extended Data Fig. 3a–c).

2. Data Interpretation: How were PCA, temporal clustering, and pathway enrichment used to identify conserved adipogenic programs?

Answer — Data Interpretation

Short answer: PCA established that component 1 is the differentiation axis; 'conserved' was defined operationally by correlated temporal profiles in all four models; supervised clustering split those profiles into early, intermediate and late programmes; and one-sided Fisher enrichment attached a pathway identity to each cluster, yielding the conserved adipogenic programme.

- **What PCA showed (Fig. 1c,d; Extended Data Fig. 2a–n):** components 1 (24.7%) and 2 (13.8%) place undifferentiated cells at one end and mature adipocytes and primary pACs at the other, so the differentiation trajectory is the dominant source of variance. Loadings identify the drivers as lipid-droplet and lipid-biosynthesis proteins (PLIN1, PLIN4, FABP4, ADIPOQ, ACSL1, MGLL, ACACB, GPD1, SCD, LIPE, ABHD5). hWA sits in a less mature position, which the authors verify against marker mRNA and protein levels, and PCAs computed within each model separately give the same first axis — so the axis is differentiation, not cell-line identity.
- **Defining the conserved set (Fig. 1e,f):** ANOVA at FDR < 1% within each model identifies 3,939 proteins that change in at least three of four models; of these, 'conserved temporal trajectory' is then defined strictly as a positive pairwise Pearson correlation of z-scored temporal profiles between every pair of models, which leaves 1,979 proteins. This two-step definition is what separates a shared programme from a change that happens to be significant in one model.
- **Temporal clustering (Fig. 1g,h):** supervised Euclidean hierarchical clustering of the z-scored profiles gives early, intermediate and late clusters. Early: cell-cycle and protein-biosynthesis proteins fall while glycosaminoglycan degradation and lysosomal proteins rise. Intermediate: fatty-acid metabolism, TCA cycle, respiratory chain and ATP synthesis rise, while DNA replication, cytoskeletal/adhesion proteins and WNT signalling fall. Late: spliceosome and mRNA-processing proteins fall while cholesterol biosynthesis and several lipid-droplet proteins rise.
- **Pathway enrichment (Fig. 1h; Methods):** one-sided Fisher's exact test against UniProtKB, GO, KEGG and CORUM annotations, with an enrichment score > 2 and Benjamini–

Hochberg FDR < 0.1, converts each cluster into a named programme; bubble size encodes enrichment and colour the P value.

- **The internal control that strengthens the interpretation (Extended Data Fig. 2p):** the 1,500 proteins that do not change in any model are enriched for transcription factors, chromatin modifiers and plasma-membrane transporters, i.e. the remodelling is selective rather than a global turnover artefact. Repeating the clustering on proteomic-ruler copy numbers recovers the same temporal modules (Extended Data Fig. 3a–c), so the programme is independent of the normalisation strategy.
- **Why this is a defensible interpretation:** because membership of the conserved set requires agreement across two donor backgrounds, immortalisation status and different differentiation protocols, the early/intermediate/late modules can be read as the core proteomic programme of human adipogenesis; model-specific clusters (Extended Data Fig. 3f,g) are then interpreted separately as age-, sex- or immortalisation-related features rather than folded into the core.

3. Multi-Omics Integration: How were transcriptomic and spatial proteomic data integrated, and what evidence established the function of C19orf12?

Answer — Multi-Omics Integration

Short answer: integration happened in two stages — abundance against the matched hAPC transcriptome (to show where mRNA fails as a proxy), then abundance against the protein-correlation-profiling (PCP) localisation map (to add space) — and the combination nominated C19orf12 as an adipocyte lipid-droplet protein, whose function was then established experimentally.

- **Transcriptome integration (Extended Data Fig. 4; Methods 'Integration of proteome and transcriptome'):** the hAPC proteome time course was matched to the previously published hAPC RNA-seq time course of the same cell model; Spearman correlations between log2 LFQ and normalised counts were 0.39–0.48 at individual time points and improved at later stages. Protein changes generally lagged mRNA changes, 16% of temporal protein profiles anti-correlated with their mRNA, and metabolic pathways correlated better than signalling, chromatin and transcriptional regulators, with protein complexes and non-membrane-bound organelles correlating worst. The integration therefore argues that post-transcriptional regulation is a major determinant and that mRNA alone cannot substitute for the proteome.
- **Spatial layer (Fig. 2; Methods 'PCP analysis'):** protein correlation profiling — mechanical lysis, continuous 20–55% sucrose gradient, 100,000 g for 3 h, 0.5 ml fractions plus the top lipid-droplet fraction, DIA LC–MS on every fraction, 0–1 scaling of profiles and removal of proteins whose replicate profiles correlate negatively. Support-vector machines (RBF kernel, C = 8, sigma = 0.2) trained on validated markers (HeLa organelle map, Protein

Atlas, Lipid Droplet Knowledge Portal) reached 96% mean accuracy in adipocytes and 88% in preadipocytes, localising 5,530 and 5,675 proteins with 4,452 and 4,912 at high confidence (Fig. 2d,e).

- **How the two layers were combined (Fig. 3, Fig. 4):** 1,323 proteins were assigned to different organelles in preadipocytes versus adipocytes (654 with high confidence in both); organelle shares shift during differentiation (mitochondria, ER, endosome and lipid droplets up; cytosol and nucleus down) and converge on the pAC profile; roughly 20% of the mapped proteome relocalises. The paper then reads abundance and localisation together: BCAT2 moves into mitochondria while BCAT1 is lost, and ALDH1L1/ALDH1L2 move from mitochondria to cytosolic complexes, which is how a compartment-specific re-routing of BCAA catabolism and one-carbon metabolism is inferred rather than assumed.
- **Nomination of C19orf12 (Fig. 5c, Fig. 6a; Extended Data Fig. 10a–c):** the lipid-droplet proteome was defined by enrichment in the lipid-droplet fraction relative to the total proteome; 59 lipid-droplet proteins were shared by the two white-adipocyte models and 34 were adipocyte-exclusive (not enriched in any of the cell-line or liver lipid-droplet datasets of the knowledge portal); intersecting this with the core proteome gave 29 significantly regulated candidates, of which C19orf12 had the most consistent temporal trajectory across all four models and the strongest mRNA and protein induction relative to SVF.
- **Functional evidence (Fig. 6b–o; Extended Data Fig. 10):** PCP predicted lipid-droplet, Golgi and mitochondrial association, and immunostaining confirmed C19orf12 around lipid droplets. Co-immunoprecipitation with C19orf12–GFP pulled down TOM40, TOM22 and TOM5 (plus GOLGA4 and C12orf75), and co-staining with TOM20 placed C19orf12 at lipid-droplet–mitochondria contact sites. siRNA knockdown increased lipid-droplet number, halved basal lipolysis and raised adiponectin secretion by ~50%; knockdown at day 8 reduced insulin-stimulated lipogenesis and basal lipolysis; Seahorse showed a lower maximal oxygen consumption rate that palmitate supplementation did not rescue and that etomoxir abolished, while maximal isoprenaline-induced lipolysis remained intact — i.e. lipases still reach the droplet and the defect lies in mitochondrial long-chain fatty-acid utilisation. In independent human WAT cohorts, C19orf12 mRNA correlated inversely with body fat, adipocyte volume, plasma triglycerides and HOMA-IR.
- **Why the integration is what made this possible:** the candidate required three layers simultaneously — an abundance time course (is it regulated during differentiation?), a localisation map (is it genuinely a lipid-droplet protein?) and the transcriptome (is the induction transcriptional?) — and each functional claim was then tested orthogonally by imaging, interaction proteomics, knockdown and bioenergetics rather than inferred from correlation.

Paper 2 — A Universal Gut-Microbiome-Derived Signature Predicts Cirrhosis

Cell Metabolism, 2020 | Metagenomics + Metabolomics + Machine Learning

1. Analysis Workflow: How were shotgun metagenomic and metabolomic data processed to identify disease-associated features?

Answer — Analysis Workflow

Short answer: shotgun metagenomes were profiled with MetaPhlAn2/HUMAnN2 and stool metabolomes by UPLC–qTOF with MS-DIAL/MS-FINDER, each reduced to a quality-controlled, normalised feature table (310 species, 7,355 pathways, 435 metabolites) before any comparison between NAFLD-cirrhosis and controls.

- **Cohort and phenotyping (Table 1; Methods 'Human subjects'):** 163 participants — 27 biopsy- or imaging-proven NAFLD-cirrhosis probands (16 compensated, 11 decompensated) and 54 non-NAFLD probands, plus 82 first-degree relatives. NAFLD was defined by AASLD criteria with MRI-PDFF ≥ 5%; cirrhosis by histology or by MRE ≥ 4.69 kPa (advanced fibrosis ≥ 3.63 kPa), with VCTE ≥ 11.8 kPa used when MRE was contraindicated. Fasting serum and stool were collected at the same visit and stored at −80 °C, so microbiome, metabolome and clinical data are time-matched.
- **Metagenomics (Methods 'Metagenome profiling'; Fig. 1A):** DNA extraction with the Qiagen MagAttract PowerSoil kit following the Earth Microbiome Project protocol, quarter-scale NexteraXT libraries, Illumina MiSeq sequencing; skewer 0.2.2 adapter trimming, Bowtie2 removal of human reads against hg19; taxonomic profiling with MetaPhlAn2 2.7.7 and functional profiling with HUMAnN2 0.11.2 (ChocoPhlAn pangenome, DIAMOND against UniRef90, MetaCyc pathway calls), renormalised with humann2_renorm_table and handled downstream through curatedMetagenomicData.
- **Yield and filtering (Results 'Taxonomic profiling'):** 1,216,335 gene families and assembled sequences for 356 species were obtained; after removing unannotated species, 310 species (53 families, 115 genera), 188 strains and 7,355 pathways remained, of which 39 pathways differed between groups. Filtering is done on the feature table, not on the machine-learning step.
- **Metabolomics (Methods 'Metabolome profiling'):** extraction solvent was benchmarked (50% MeOH, 80% EtOH, water and EtOAc:MeOH:H2O), and 50% MeOH was chosen on the basis of base-peak complexity; 5 µl was separated on a Kinetex C18 column by UPLC and ionised into a qTOF (Bruker MaXis) with DDA top-10 fragmentation (35 eV for +1, 25 eV for +2 ions), 30-s dynamic exclusion and lock mass calibration. MS-DIAL 3.66 performed feature extraction, alignment and putative annotation (MS1 0.01 Da, MS2 0.025 Da, RT tolerance 0.2 min) with manual false-positive screening, and MS-FINDER 3.20 assigned identities at ≤ 10 ppm mass error and MS2 score ≥ 8, yielding 435 metabolites.
- **Comparative statistics (Methods 'Statistical analysis'):** alpha diversity by Inverse Simpson (phyloseq), beta diversity by weighted-UniFrac PCoA with PERMANOVA

(adonis), differential abundance with DESeq2, zero-inflated counts with a fast zero-inflated negative binomial mixed model, species–metabolite association by Spearman correlation, and pathway-set reproducibility across cohorts by the RV coefficient (FactoMineR).

- **Why the order matters:** every data type is reduced to a feature table with an explicit QC, filtering and normalisation record before any modelling, which is what allows a 19-species and a 17-metabolite model to be compared on the same footing and then transferred unchanged to cohorts generated by different sequencing technologies.

2. Predictive Modeling: How were microbial and metabolite signatures selected, modeled, and validated using random forests and ROC/AUC?

Answer — Predictive Modeling

Short answer: features were nominated by differential abundance (DESeq2) and ranked by random-forest Gini importance; compact signatures (19 species, 17 metabolites) were trained with 10-fold cross-validation, evaluated by ROC/AUC, then frozen and applied unchanged to independent cohorts and to a mixed-fibrosis cohort.

- **Feature selection (Fig. 1E, Fig. 2A, Fig. 3A):** differentially abundant features were first identified with DESeq2; random forest (R caret) then ranked features by mean decrease in Gini, and the top-performing model defined the signature. This two-filter design avoids selecting features on the same statistic that is later used for evaluation, and keeps the signature small (19 species out of 310) relative to the sample size.
- **Training and internal evaluation (Fig. 1F, Fig. 3B,C):** random forest was trained with caret's train() and trainControl() using 10-fold cross-validation; ROC curves and AUCs were computed with pROC. The species model achieved AUC 0.91 (95% CI 0.89–0.93) and the metabolite model the same AUC 0.91; PLS/OPLS scores (ropls) provided an independent multivariate confirmation that the 17 metabolites separate the groups (Fig. 3C).
- **What the signature contains (Fig. 1E):** enriched in cirrhosis: Veillonella parvula (the most discriminatory adverse species despite low abundance), V. atypica, Ruminococcus gnavus, Clostridium bolteae and Acidaminococcus sp. D21 (15-fold); depleted: Eubacterium eligens, Eubacterium rectale and Faecalibacterium prausnitzii — the last being the most important protective species in the model, consistent with its anti-inflammatory role and its loss in other intestinal disorders.
- **External validation without retraining (Fig. 5A–D):** because the preprocessing was fixed a priori (z-score centre and scale on the relative-abundance tables), the trained model could be applied as-is: Italian 16S cohort AUC 0.89; Chinese metagenomic cohort AUC 0.86, comparable to the AUC 0.84 of that study's own 15-marker signature; UCSD relatives AUC 0.88 with 19 species + age; and with serum albumin — the only clinical variable available in the Chinese cohort — a 22-feature model reached AUC 0.91 (relatives) and 0.95 (Chinese).

- **Cirrhosis versus fibrosis (Fig. 5E–H, I,J):** in the independent biopsy cohort (36 F0, 41 F1–3, 9 F4), species + age separated cirrhosis from NAFL with AUC 0.85 and from F1–3 with AUC 0.84; adding AST, which rises in cirrhosis, raised these to 0.94 and 0.91. Kruskal–Wallis, FZINBMM and DESeq2 confirmed stage-dependent shifts in the abundance of individual signature species across NAFL, mild/moderate fibrosis and cirrhosis.
- **Why the validation is persuasive:** the test cohorts differ in geography, aetiology (alcohol, HBV, HCV, NAFLD), ancestry and even sequencing chemistry (16S versus shotgun), which is exactly the heterogeneity an epidemiologically 'universal' signature must survive; performance is reported with confidence intervals rather than as a point estimate, and the study's power was justified in advance from an earlier 16-versus-33 cohort (≥ 80% power at $\alpha = 0.01$).

3. Multi-Omics Integration: How were microbial pathways, metabolites, and clinical parameters integrated to identify cirrhosis-associated signatures?

Answer — Multi-Omics Integration

Short answer: species, pathway and metabolite tables were linked by correlation networks and by a single-microbe culture experiment, then combined with three routine clinical variables (age, albumin, AST) into staged composite models that were re-tested in independent cohorts.

- **Species–metabolite association (Fig. 4A):** a Spearman correlation matrix of the 19 discriminatory species against the 17 discriminatory metabolites showed specific pairings — C18-sphingosine with Ruminococcus gnavus, kynurenic acid with Faecalibacterium prausnitzii, tryptophan/indole derivatives with taxa of the up-signature.
- **Extended network (Fig. 4B):** HUMAnN2 gene families, species and metabolites were assembled into a 33-node, 338-edge network ($P < 0.05$, Cytoscape), which asks whether the discriminating taxa actually carry the enzymatic potential to produce the discriminating metabolites — the bridge from 16S-level or metagenomic taxonomy to chemistry.
- **Experimental check of one edge (Fig. 4C):** Ruminococcus gnavus VPI C7-9 was grown anaerobically with a chenodeoxycholic acid gradient (0.001–5,000 μg ml-1), approximating the raised bile-acid levels of cirrhosis, and LC–MS/MS showed dose-dependent increases in C18-sphingosine and 1H-indole-3-carboxaldehyde — converting a correlation into a plausible production mechanism.
- **Pathway-level concordance across cohorts (Fig. 2B):** aromatic and branched-chain amino-acid biosynthesis pathways discriminate the groups, track Acidaminococcus and Veillonella, correlate with clinical metadata, and reproduce across the Chinese cohort and the earlier San Diego cohort (RV coefficients, $P = 6.1 \times 10^{-6}$ for the top pathways; $P = 0.0051–0.0052$ for the full pathway set).
- **Clinical integration (Fig. 5C,D,G,H; Fig. S6):** the 19-species model was augmented by age, then by albumin (22 features, testable in both cohorts) and by AST. The result is a

staged composite: the microbiome component answers ‘is this cirrhosis?’ across aetiologies, while AST answers ‘is it worse than fibrosis?’ within an aetiology-specific cohort.

- **The actual integrated claim:** the same taxa (Veillonella, Streptococcus, Acidaminococcus), the same pathways (AAA/BCAA biosynthesis) and the same metabolite classes (tryptophan/indole derivatives, BCAA derivatives, bile acids, vitamin D) emerge from three independent data types while alpha diversity falls in parallel — that convergence, rather than any single AUC, is what supports a cirrhosis-associated gut signature; and the authors state plainly that association is not causality.

Paper 3 — Shared and Disease-Specific Host Gene–Microbiome Associations

Nature Microbiology, 2022 | Transcriptomics + Microbiomics

1. Analysis Workflow: How were paired host RNA-seq and 16S rRNA sequencing data processed and compared across CRC, IBD, and IBS?

Answer — Analysis Workflow

Short answer: paired mucosal RNA-seq and 16S tables from three cohorts were each processed to a common analytical shape (protein-coding genes after VST; CLR-transformed taxa concatenated across taxonomic ranks) and then compared by Procrustes/Mantel before any integration.

- **Cohorts (Fig. 1a; Supplementary Table 1):** 208 paired host transcriptome/microbiome samples (416 samples in total) — CRC 88 pairs (44 tumours and 44 patient-matched normals), IBD 78 pairs (56 patients: 34 Crohn's disease, 22 ulcerative colitis; plus 22 non-IBD controls from HMP2) and IBS 42 pairs (29 IBS, 13 controls). Every individual contributes one microbiome and one host-expression sample from the same colonic mucosal biopsy, which is the precondition for any association claim.
- **Host RNA-seq (Methods ‘Host RNA-seq sequencing, alignment and quality control’):** for the newly generated CRC data, RNA was checked for yield and RIN ≥ 8, converted with Illumina TruSeq stranded total RNA chemistry and sequenced on HiSeq 2500; FastQC 0.11.5, Trimmomatic trimming (3-bp sliding window, Q16), HISAT2 alignment to GRCh38 (mean 87.11%, 14.4–31.5 million aligned reads), then featureCounts (Subread 1.4.6). IBD data came through the Broad Picard pipeline and IBS data through MAP-RSeq 2.0.0/TopHat 2.0.12 on hg19.
- **Unified host preprocessing (Methods ‘Preprocessing host gene expression data’):** biomaRt restricted features to protein-coding genes; genes had to be expressed in at least half of the samples; counts were variance-stabilised with DESeq2 1.14.1 and genes below the 25% variance quantile were dropped, leaving 12,513 (CRC), 11,985 (IBD) and 12,429 (IBS) genes. Filtering is therefore per cohort, not pooled.

- **Microbiome processing (Methods 'Preprocessing microbiome data'):** CRC used V5–V6 amplicons (MiSeq 2 × 250, USEARCH v7, QIIME 1.7.0 OTUs), IBD and IBS used V4 (UPARSE; Shi7 with BURST). Per cohort, archaea, chloroplast, kit and environmental contaminant sequences were removed; taxa were summarised at six ranks; taxa at ≥ 0.001 relative abundance in ≥ 10% of samples were kept; the six rank tables were concatenated and CLR-transformed together to respect compositionality (the reverse transformation order gave Pearson r = 0.92, Spearman rho = 0.87, so the choice is immaterial) — 235 taxa in CRC, 121 in IBD and 238 in IBS, with the lower IBD diversity matching HMP2 reports.
- **Comparing the two data types (Methods 'Procrustes analysis and Mantel test'):** Procrustes analysis used Aitchison distance for host expression and Bray–Curtis for microbiome composition, with 9,999 permutations, and was verified by Mantel tests. Global correspondence was significant only in CRC (Procrustes P = 1 × 10^-4; Mantel P = 0.0026) and absent in IBD (P = 0.1; 0.2597) and IBS (P = 0.42; 0.9525).
- **Batch and design control (Methods 'Integrating host gene expression and gut microbiome data'):** because protocols differ between cohorts, the study uses a meta-analysis design — integration and case–control comparison are done within each cohort and results are combined only at the final step; OTU versus ASV (DADA2) profiles were compared (phylum rho = 0.89, class 0.68, genus 0.6) and sample-size/power differences were handled by differential enrichment rather than by pooling. The workflow's conclusion is quantitative: the two layers are not globally coupled, so integration has to be sparse and targeted — which motivates the algorithms used next.

2. Integration Algorithms: How were sparse CCA, LASSO regression, and pathway enrichment used to identify host–microbe associations?

Answer — Integration Algorithms

Short answer: sparse canonical correlation analysis produced group-level components (host gene sets against microbial sets) with L1 penalties tuned by leave-one-out cross-validation; gene-wise lasso regression with desparsified inference and stability selection produced individual gene–taxon associations; Fisher enrichment translated both into pathways.

- **Sparse CCA (Methods 'Sparse CCA'):** the objective maximises u^T X^T Y v subject to $||u||1 \leq \lambda 1$ and $||v||1 \leq \lambda 2$, so the L1 constraints perform feature selection while the canonical correlation captures joint variation — the standard answer to high-dimensionality and multicollinearity (implemented with PMA 1.1). Penalties were tuned by a grid search with leave-one-out cross-validation: $\lambda 1/\lambda 2$ = 0.15/0.2 (CRC), 0.177/0.333 (IBD) and 0.4/0.1 (IBS).
- **Component significance and yield (Methods 'Significance of correlation for sparse CCA components'):** ten components were computed per cohort, for cases and controls separately, iteratively so that components are uncorrelated; significance was obtained from held-out

scores (leave-one-out) with cor.test and Benjamini–Hochberg FDR < 0.1, giving 7 significant components in CRC (on average 828 host genes and 8 microbes), 4 in IBD (2,095 genes, 6 microbes) and 6 in IBS (577 genes, 61 microbes). Significance is tested at component level, not per feature.

- **Keeping only case-specific signal (Methods 'Comparing case versus control associations and pathways'):** associations present in controls were discarded; where a pathway was enriched in both, a comparative log-odds-ratio statistic, $z = [\log \delta(\text{case}) - \log \delta(\text{control})]/SE$, was used to retain pathways differentially enriched in cases (FDR < 0.2). No gene–microbe association overlapped between cases and controls in any cohort, and downsampling cases to control numbers did not create any, so case-specificity is not an artefact of sample size.
- **Lasso regression (Methods 'Lasso regression analysis'):** one model per host gene, with gene expression as the response and all taxon abundances plus covariates (sex; Crohn's/UC; IBS-C/IBS-D) as predictors (glmnet 2.0-13, coordinate descent, λ by leave-one-out CV). Because lasso coefficients are biased, inference used the desparsified lasso (hdi 0.1-7) to obtain 95% confidence intervals and P values, corrected by Benjamini–Hochberg FDR < 0.1.
- **Stability selection (Methods 'Stability selection for the lasso model'):** n/2 subsampling, 100 refits with a randomly perturbed λ, and retention of taxa selected in at least 60% of fits (stabs 0.6-3), intersected with the FDR-significant set. This is what makes the 755 (CRC: 745 genes, 120 microbes), 1,295 (IBD: 1,246 genes, 56 microbes) and 441 (IBS: 436 genes, 102 microbes) reported pairs robust, and it also protects against the correlated features created by concatenating taxonomic ranks into one matrix.
- **Pathway enrichment (Methods 'Enrichment analysis'):** one-sided Fisher's exact test with the actually modelled genes as background, KEGG/PID for sparse CCA and KEGG/PID/REACTOME for lasso, pathway-size windows of 25–300 genes (CCA) or 10–85 genes (lasso) with at least five overlapping genes, and BH FDR < 0.1 — giving shared pathways (oxidative phosphorylation, RAC1, focal adhesion, proteasome) plus 102 disease-specific pathways (52 CRC, 25 IBD, 25 IBS) at component level and 18 unique pathways (4/9/5) at gene level. More than 12,300 gene-wise models per cohort were run in parallel on five Minnesota Supercomputing Institute nodes (~5.5 h per cohort) and networks were drawn in Cytoscape 3.5.1 with edge width = Spearman rho and colour = sign.

3. Biological Interpretation: How were shared and disease-specific regulatory networks identified, and what biological conclusions can be drawn?

Answer — Biological Interpretation

Short answer: three complementary comparisons — shared pathways with disease-specific microbes, shared microbes with disease-specific host genes, and shared host genes with different

microbes — show that disease-specificity lies in the pairing, which supports a model in which a common dysbiotic core is transduced through disease-specific host programmes.

- **Shared pathways, different microbes (Fig. 2b):** RAC1 signalling is microbiome-associated in all three diseases, yet its microbial partners differ: Streptococcus, Synergistales and GN02 in CRC; Granulicatella and Clostridium sensu stricto 1 in IBD; Bacteroides massiliensis, Bifidobacterium and Odoribacter in IBS. The same pattern holds for oxidative phosphorylation and for the adhesion/integrin modules, which is why the authors call the pathways 'shared' and the taxa 'disease-specific'.
- **Disease-specific pathways (Fig. 2c):** CRC: Syndecan-1, paired with Parvimonas and Bacteroides fragilis (both known colorectal carcinogenesis taxa); IBD: integrin beta-1 and A6B1/A6B4 pathways, paired with Peptostreptococcaceae, Intestinibacter and Phascolarctobacterium; IBS: ribosome, B-cell receptor signalling and immune-response pathways.
- **Common host genes (Fig. 2d):** the genes shared between components across diseases are enriched for VEGF signalling, complement and coagulation cascades and cytokine–cytokine receptor interaction — i.e. the shared element is host epithelial immunovascular biology, not a shared microbe.
- **Shared microbes, different host genes (Fig. 4a):** Peptostreptococcaceae, Streptococcus and Staphylococcus associate with host genes in all three diseases, but the genes differ: Peptostreptococcaceae links positively to PYGB and NCK2 in IBD, negatively to HAS2 in IBS and negatively to GAB1 in CRC; Streptococcus links to RIPK4 (WNT/NF-κB) in CRC but to DPEP2 (macrophage inflammatory response) in IBS. Blautia is negatively associated with RIPK3 in both CRC and IBD, and with ZBP1 in CRC — a butyrate producer tracking an inflammation/necroptosis axis.
- **Shared host genes, different microbes (Fig. 4b):** 135 genes are associated in at least two diseases. PINK1 (mitochondrial homeostasis, PI3K/AKT) pairs with Collinsella in CRC, Peptostreptococcaceae in IBD and Blautia in IBS; ANXA1 is positive with Bacteroidales in CRC but negative with Peptostreptococcaceae in IBD; TLR4 pairs with GN02 in CRC and Acidaminococcaceae in IBD.
- **What can be concluded, and what cannot:** the regulatory logic identified is that the same taxon is associated with different host genes, and the same host pathway with different taxa, in each disease — consistent with the mucosal context (transformed epithelium, inflammatory milieu) conditioning the interaction. The conclusion is explicitly bounded: the analysis is correlational and cannot establish directionality, only taxonomic composition was profiled (no microbial gene or pathway data), and age, diet, host genetics, medication and sampling site are uncontrolled across cohorts. The networks are therefore presented as testable hypotheses for cell and animal work, not as causal pathways.

Paper 4 — α-Ketoglutarate Signaling to Chromatin

Science, 2026 | Functional Genomics + Metabolomics + Epigenomics + Transcriptomics

1. Experimental Workflow: How was the αKG biosensor developed and integrated with CRISPR screening to identify metabolic regulators?
2. Multi-Omics Analysis: How were metabolomic, ChIP-seq, and transcriptomic results analyzed to connect αKG metabolism with chromatin regulation?
3. Mechanistic Validation: How did genetic perturbation, epigenomic profiling, and αKG rescue experiments establish the GPT2–SLC25A11 regulatory mechanism?

Note: Paper 4 was not attempted. The assignment requires any three of the four papers; Papers 1, 2 and 3 are answered above, and Paper 4 questions are left as printed in the template.